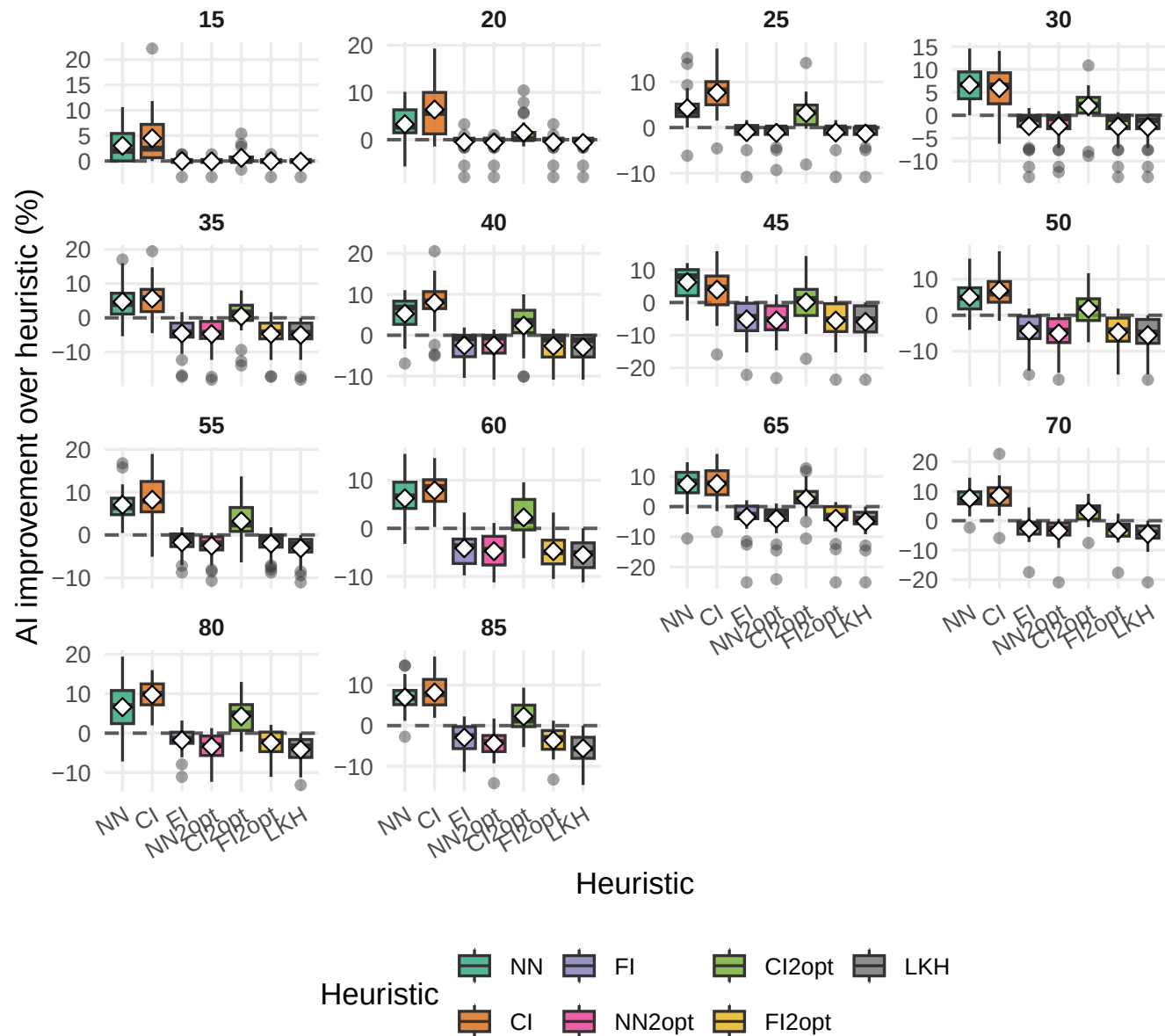


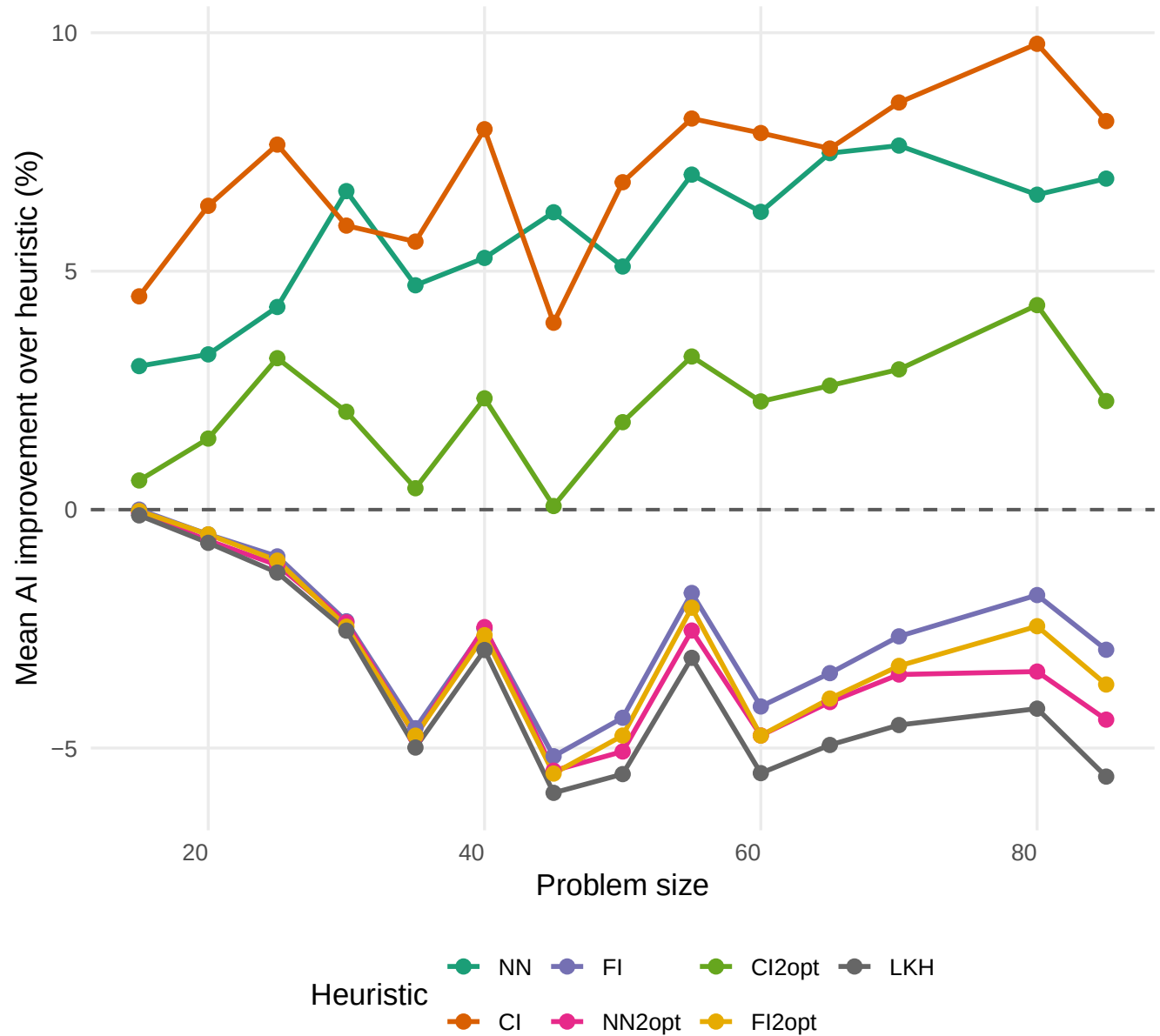
Distribution of AI Improvement over Heuristics

Boxplots show variation across runs; white diamonds show the mean



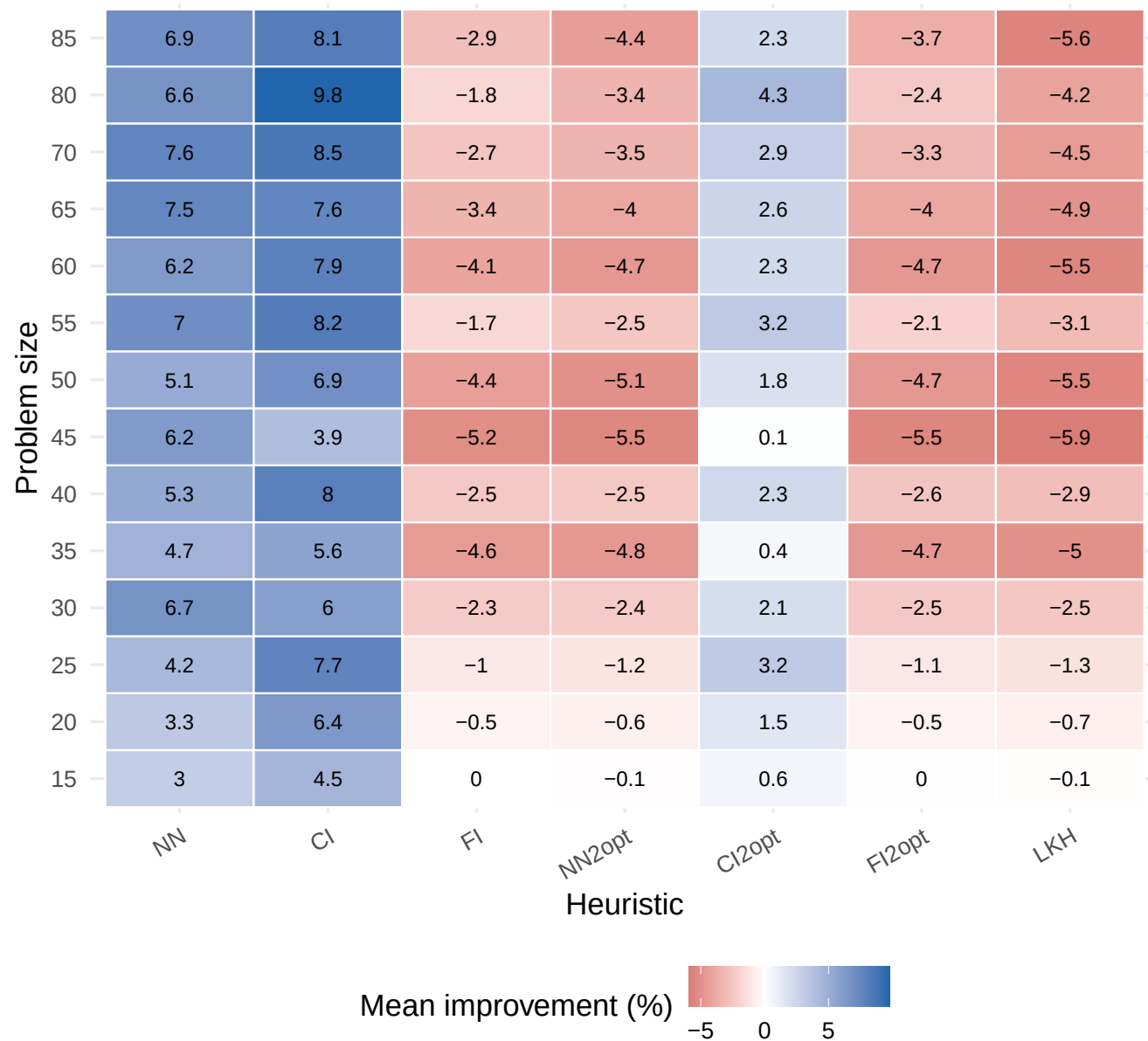
Mean AI Improvement by Problem Size

Positive values indicate that AI is better than the heuristic on average



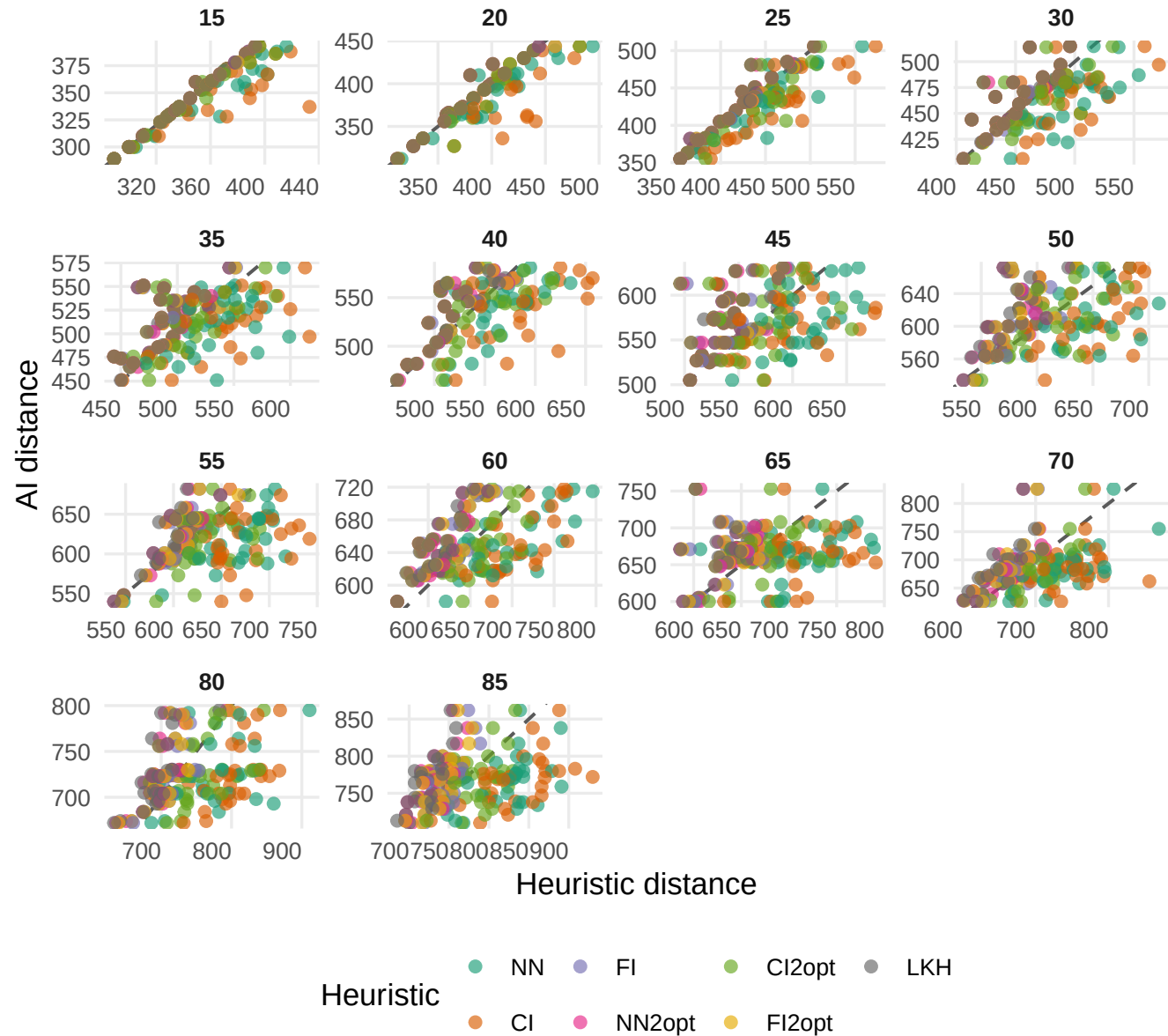
Mean AI Improvement Heatmap

Blue = AI better, red = AI worse, white = approximately equal



AI Distance vs Heuristic Distance

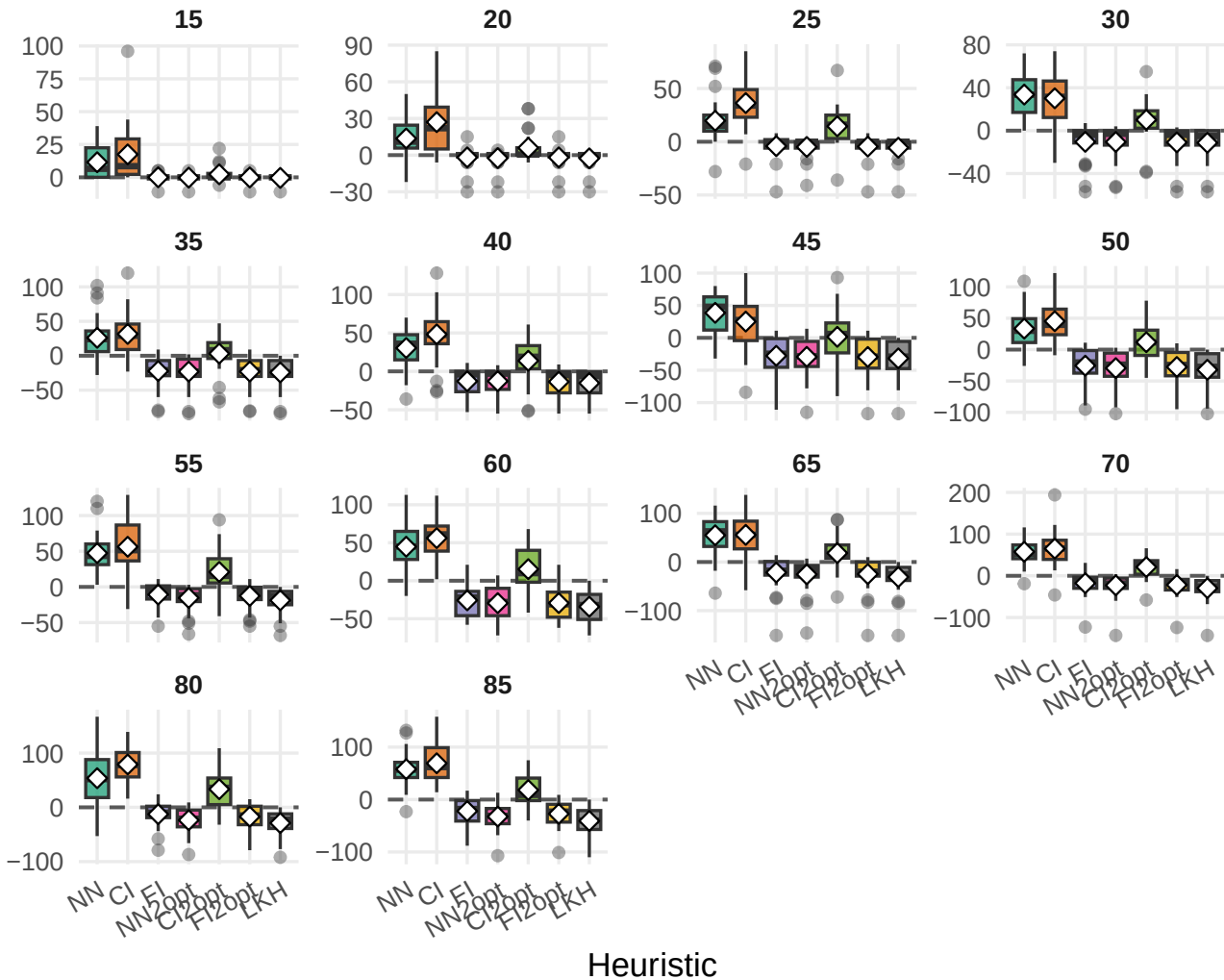
Points below the dashed line indicate that AI found a shorter route



Absolute Distance Difference: AI vs Heuristic

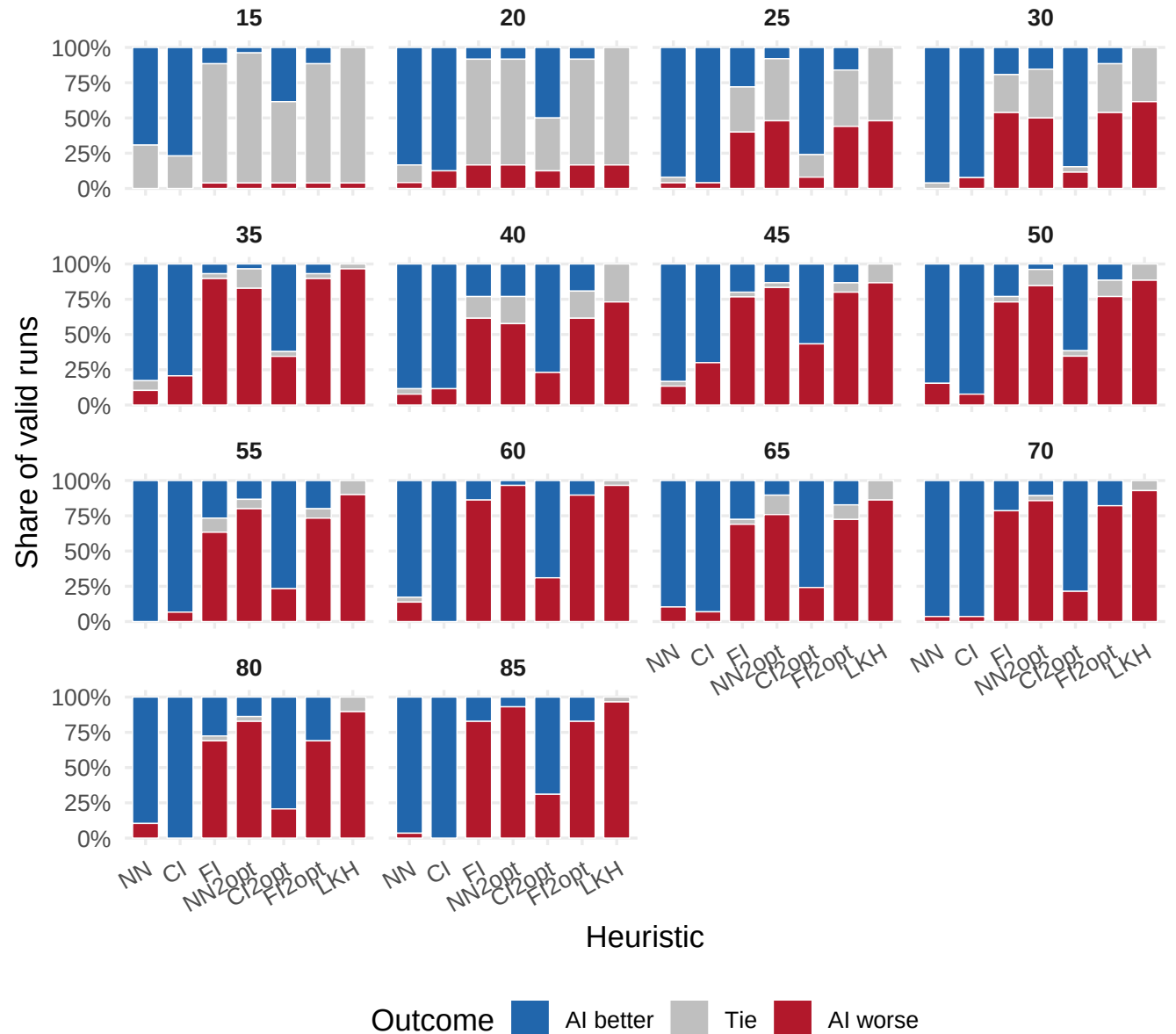
Positive values mean AI found a shorter route; white diamonds show the mean

Heuristic distance - AI distance



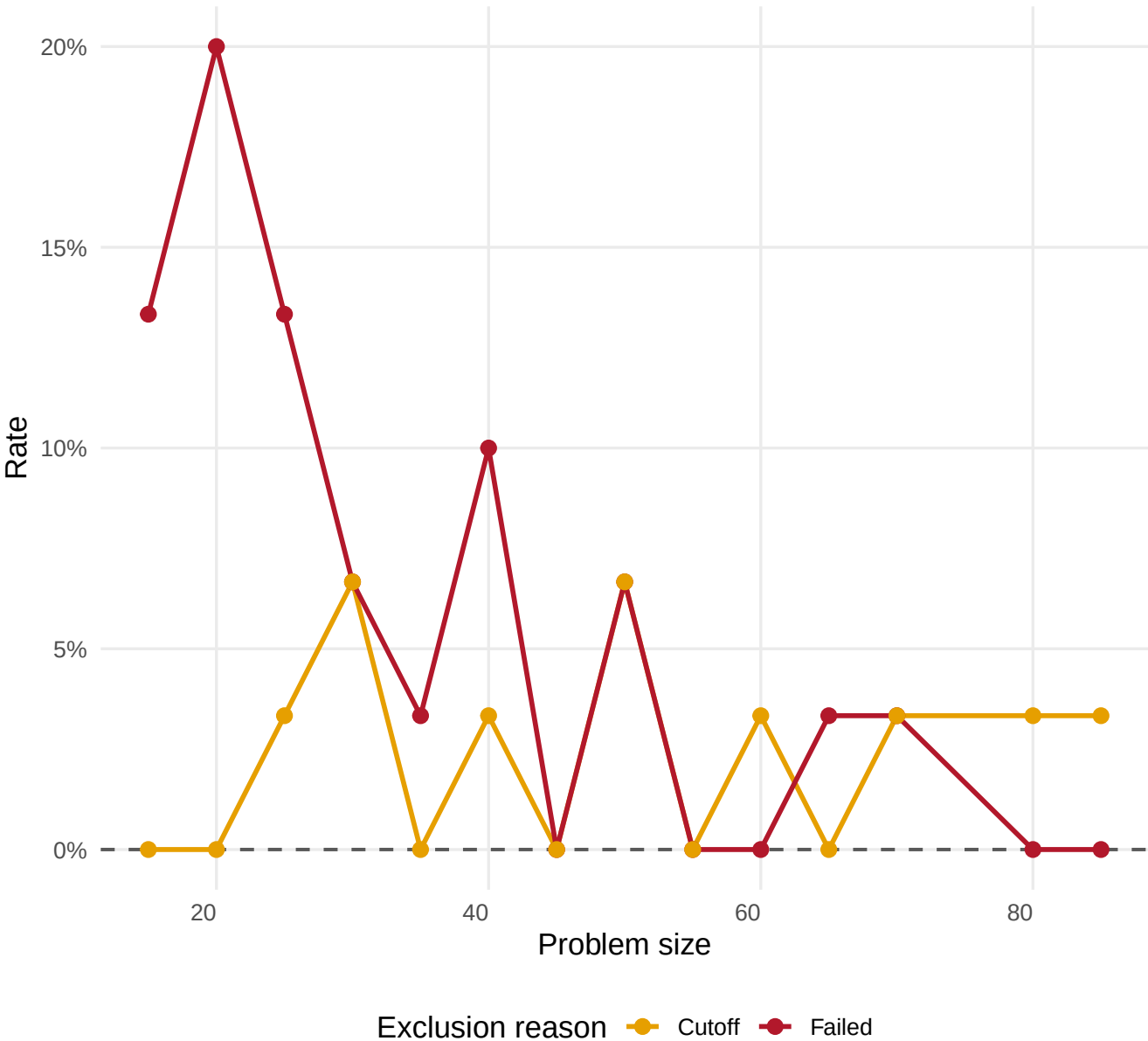
AI Win / Tie / Loss Rate by Heuristic

Based on improvement percentage for each valid run



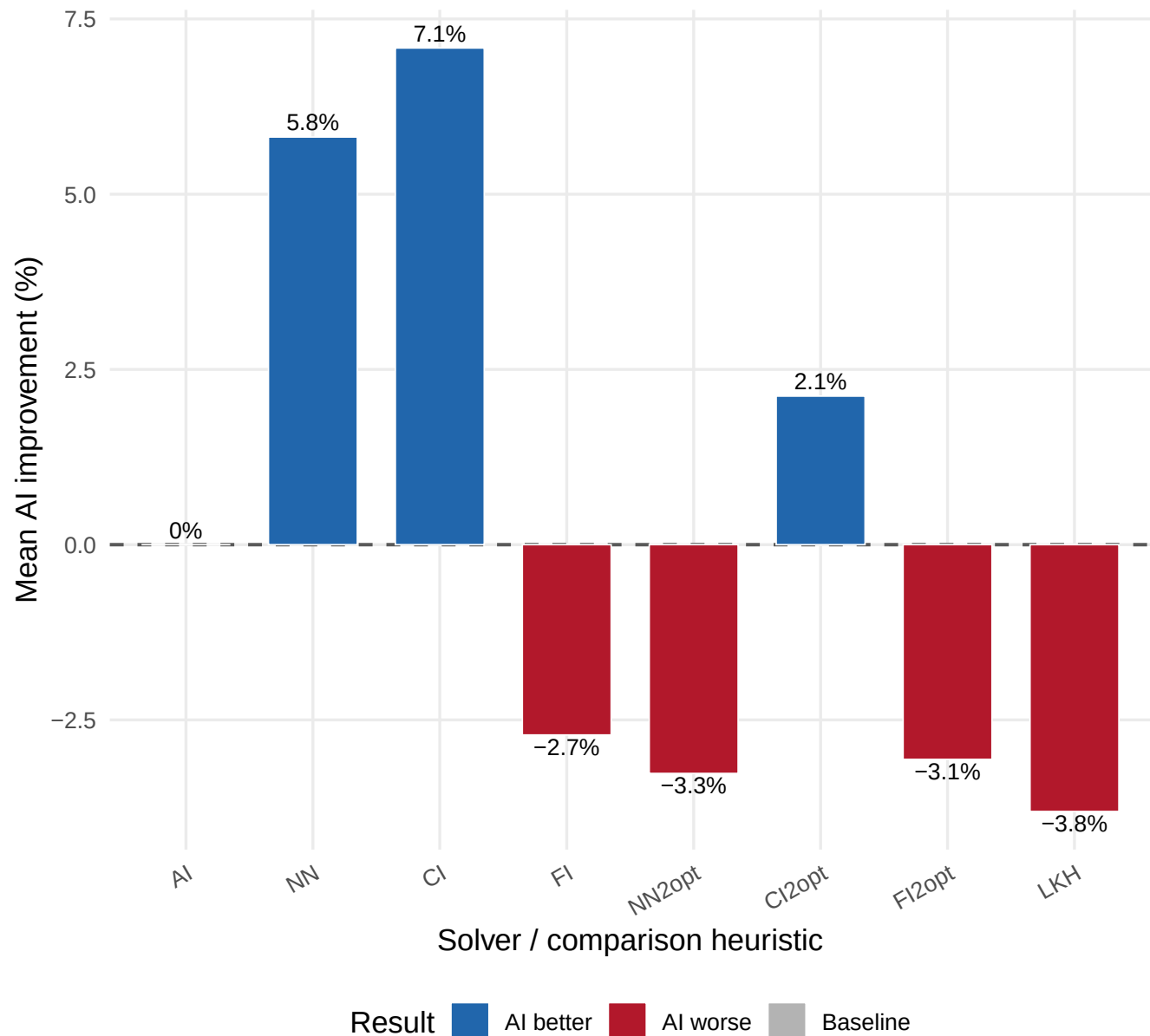
Failure and Cutoff Rate by Problem Size

Each run is counted once, not once per heuristic



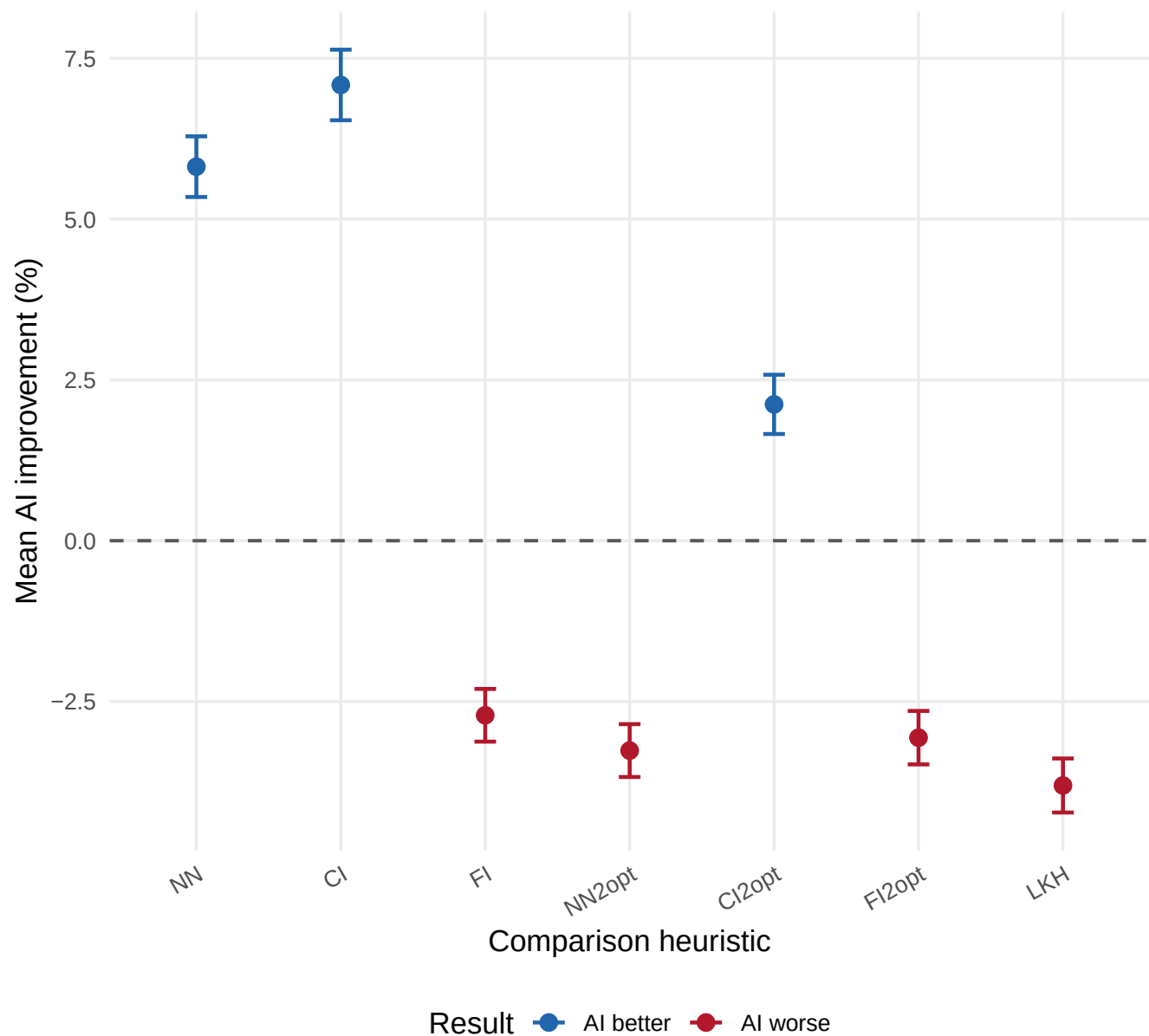
Overall Mean AI Improvement over Heuristics

Positive values mean AI is better on average



Confidence Intervals for Overall Mean AI Improvement

Points show means; vertical lines show approximate 95% confidence intervals



Confidence Intervals for Mean AI Improvement by Problem Size

Points show means; vertical lines show t-based 95% confidence intervals

